

Optical Path Length (OPL) Drift Modelling and Feedback Control

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1 Introduction

In this report I present the formulation of my OPL drift model and its application to temperature feedback control in the spectrograph system. The model quantifies how variations in environmental parameters (temperature, pressure, humidity) affect the measured OPL, and provides a way to correct for these effects using active thermal feedback.

2 Mathematical Model

Let the measured OPL deviation at time t be

$$y(t) = \Delta\text{OPL}(t).$$

I model $y(t)$ as a linear time-invariant system with environmental inputs:

$$\hat{y}(t) = a_0 + a_1 t + \sum_{k \in \mathcal{K}} w_k (u_k * g_{\tau_k, \theta_k})(t),$$

where a_0 and a_1 are baseline offset and slope, w_k are channel weights, u_k are input signals, and $*$ denotes convolution. The input channels are

$$u_k \in \{\Delta T^+(t), \Delta T^-(t), \Delta P(t), \Delta H(t)\},$$

with

$$\Delta T^+(t) = \max(\Delta T(t), 0), \quad \Delta T^-(t) = \min(\Delta T(t), 0).$$

The kernel is

$$g_{\tau,\theta}(t) = \frac{1}{\tau} e^{-(t-\theta)/\tau} \mathbf{1}\{t \geq \theta\},$$

which has unit DC gain. Therefore each w_k is a DC sensitivity.

2.1 Expanded Form

Expanding by channel, I write:

$$\begin{aligned} \hat{y}(t) = & a_0 + a_1 t + \sum_{m=1}^{M_{T^+}} w_m^{T^+} (\Delta T^+ * g_{\tau_m^{T^+}, \theta_m^{T^+}})(t) + \sum_{m=1}^{M_{T^-}} w_m^{T^-} (\Delta T^- * g_{\tau_m^{T^-}, \theta_m^{T^-}})(t) \\ & + \sum_{m=1}^{M_P} w_m^P (\Delta P * g_{\tau_m^P, \theta_m^P})(t) + \sum_{m=1}^{M_H} w_m^H (\Delta H * g_{\tau_m^H, \theta_m^H})(t). \end{aligned}$$

Each convolution is

$$(u * g_{\tau,\theta})(t) = \int_0^t u(\xi) g_{\tau,\theta}(t - \xi) d\xi.$$

2.2 Channel Gains

Because $\int g_{\tau,\theta}(t) dt = 1$, the DC gain of each channel is the sum of its weights:

$$g_T = \sum_{m=1}^{M_{T^+}} w_m^{T^+} + \sum_{m=1}^{M_{T^-}} w_m^{T^-}, \quad g_P = \sum_{m=1}^{M_P} w_m^P, \quad g_H = \sum_{m=1}^{M_H} w_m^H.$$

Here:

- g_T is the steady-state OPL sensitivity to temperature,
- g_P is the sensitivity to pressure,
- g_H is the sensitivity to humidity.

I compute these by fitting weights w_k with L1-regularised regression (to select kernels), then refitting them with OLS.

3 Frozen Model

On new data files I hold $\{(w_k, \tau_k, \theta_k)\}$ fixed and only re-estimate (a_0, a_1) to account for baseline offsets. I call this the *frozen model*. This ensures physical channel sensitivities remain consistent.

4 Discrete-time Implementation

With sampling period Δt and $t_n = n\Delta t$, the model becomes

$$\hat{y}[n] = a_0 + a_1(n\Delta t) + \sum_{m=1}^{M_{T+}} w_m^{T+} z_m^{T+}[n] + \sum_{m=1}^{M_{T-}} w_m^{T-} z_m^{T-}[n] + \sum_{m=1}^{M_P} w_m^P z_m^P[n] + \sum_{m=1}^{M_H} w_m^H z_m^H[n],$$

where each $z_m^c[n]$ is a delayed exponential smoothing of its channel input.

5 Error Metrics

I quantify model accuracy using:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{N} \sum_i |y_i - \hat{y}_i|,$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

6 Feedback Control Law

To compute a temperature correction I remove contributions from pressure and humidity:

$$\Delta T_{\text{cmd}}(t) = -\frac{\Delta \text{OPL}(t) - g_P \Delta P(t) - g_H \Delta H(t)}{g_T}.$$

Here:

- g_T scales the command relative to OPL–temperature sensitivity,
- g_P and g_H subtract pressure/humidity contributions,
- the negative sign ensures feedback opposes the drift.

In practice I implement this with a PI controller, per-step clamps (e.g. ± 300 mK), and anti-windup, acknowledging a thermal lag of ~ 900 s.

7 Example Gains

From one representative fit I obtain:

$$g_T \approx -8.22 \times 10^3 \mu\text{m}/^\circ\text{C}, \quad g_P \approx 0.53 \mu\text{m}/\text{hPa}, \quad g_H \approx -0.21 \mu\text{m}/\% \text{RH}.$$

These show temperature dominates OPL drift, while pressure and humidity provide smaller but non-negligible contributions.